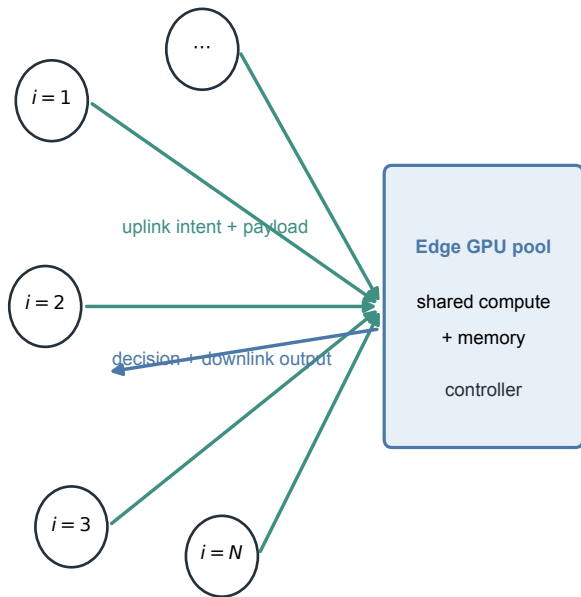


a

## Wireless edge offloading system



Each node chooses  $s_i \in \{0, 1\}$

b

## Coupled state and hard constraints

$$K(\mathbf{s}) = \sum_i s_i$$

offloaded task count

$$W(\mathbf{s}) = \sum_i s_i q_i$$

aggregate workload

$$V(\mathbf{s}) = \sum_i s_i v_i$$

aggregate memory

**Feasible:**  $W \leq W_{\max}$  and  $V \leq V_{\max}$

Congestion:  $D_{\text{comp}} = D_0 + a \rho / (1 - \rho)$ ,  $\rho = W/C_W$

WA-MCBBR accepts only feasible single flips that reduce the public objective  $J$ .